MILITARY SPECIFICATION

ANTENNA, FIXED HIGH FREQUENCY, VERTICAL LOG PERIODIC (DIPOLE)

1. SCOPE

1.1 Scope. This specification covers requirements for vertically polarized, log periodic (dipole) antennas capable of producing directive, low elevation angle electromagnetic radiation in the high frequency region of the radio spectrum (hereinafter referred to as the antenna).

1.2 Classification. The antenna shall be of the following types, classes, and configurations, as specified (see 6.2).

Type	Covers the frequency range of:
I	2.0 MHz to 30 MHz
II	2.5 MHz to 30 MHz
III	3.0 MHz to 30 MHz
IV	3.5 MHz to 30 MHz
V	4.0 MHz to 30 MHz
VI	4.5 MHz to 30 MHz
VII	5.0 MHz to 30 MHz
VIII	5.5 MHz to 30 MHz
IX	6.0 MHz to 30 MHz

Class	Wind and Ice conditions
A	100 mph (no ice)
B	125 mph (no ice), or 90 mph with 1/2 inch of radial ice on all members
C	160 mph (no ice), or 90 mph with 1/2 inch of radial ice on all members
D	Special case, as defined in the procurement document

Configuration	
Single	Single array of vertical dipole elements
Quad	Four independent orthogonal arrays of vertical dipole elements with a common central tower.

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2. APPLICABLE DOCUMENTS

2.1 The following document of the issue in effect on date of invitation for bids or request for proposal, forms a part of the specification to the extent specified herein. (In the event of conflict between the referenced documents and this specification, the latter shall govern.)

SPECIFICATION

MILITARY

MIL-A-28768 - Antennas, Fixed High Frequency, General Specification for

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

B105-55 - Hard Drawn Copper Alloy Wires For Electrical Conductors

(Applications for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pa. 19103.)

3. REQUIREMENTS

3.1 General. The antenna shall comply with the requirements of MIL-A-28768 and the requirements specified herein. Unless otherwise specified herein, requirements shall be met individually, simultaneously, and under all possible combinations. The antenna shall be furnished as a complete kit, ready for installation and operation. The antenna shall be designed in accordance with log periodic theory, using transposed half wave dipoles as the radiating elements.

3.1.1 Configuration. The basic configuration of this antenna shall be an array of parallel, vertical or near vertical half wave dipoles, sized and arranged in accordance with log periodic antenna design principles so as to provide frequency independent radiation in the direction of the shorter elements (that is, the front of the array). The length of successive dipoles shall form a geometric progression of ratio tau (τ). This ratio need not be constant, but shall never be less than 0.8. The number of active dipoles between the frequency limits of the antenna type specified shall be determined by performance requirements but shall never be less than those shown in Table I.

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Table I - Minimum Dipole elements by antenna type.

Antenna Type (See 1.2)	Minimum number of dipoles
I	15
II	14
III	13
IV	12
V	11
VI	10
VII	10
VIII	9
IX	9

The dipoles shall be suspended from an upper cable called a catenary, which in turn is suspended between a metal tower at the tall end (that is, rear) and a wooden pole at the short end (that is, front) of the array. (This array is sometimes referred to as an antenna "curtain"). When a single configuration is specified in the procurement document, a single array as described above shall be provided. When a quad configuration is specified, four arrays as described above, but with a common central tower, shall be supplied such that the angle subtended between vertical planes containing adjacent arrays is 90 degrees.

3.1.2 Materials. Materials used in the antenna shall comply with MIL-A-28768. A wooden pole shall be supplied for supporting the front of each array and a tower shall be supplied for supporting the rear. The radiating elements shall be constructed from salt spray resistant aluminum wire, aluminum clad wire, bronze wire, or corrosion resisting metal. Bronze wire shall conform to ASTM B105-55, while the other wires specified herein shall conform to MIL-A-28768.

3.2 Content of requirements. Requirements for the antenna are contained herein as follows:

Requirement	Paragraph
General	3.1
Size	3.3
Environmental conditions	3.4
Structural design	3.5
Electrical performance	3.6
Ground System	3.7

3.3 Size. The installed antenna, including tower lights and anchors under worst case ground slope conditions, shall be within the size limitations listed in Table II. All dimensions are in wavelengths at the low frequency limit of the type specified.

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TABLE II - Maximum dimensions in wavelengths.

Configuration	Height	Length	Width
Single	0.80	1.8	1.25
Quad	0.85	2.4	2.4

3.4 Environmental conditions. The antenna shall meet the environmental requirements specified in MIL-A-28768. The antenna shall be designed without the use of structural relief devices ("fuses") to meet the allowable stress requirements specified in MIL-A-28768 when loaded with the wind and ice conditions defined by the Class specified (see 1.2). Unless otherwise specified in the procurement document, Class B shall apply (see 6.2). The word "or" in the Wind and Ice condition column for each Class specified means that the antenna shall meet both conditions independently.

3.5 Structural design.

3.5.1 Field adjustment of cable tensions. The design of the antenna shall provide facilities for field adjustment of the antenna array and guy cables both during and after erection. Catenary and guy cable tensioning devices shall be so arranged that tension adjustment can be performed from the ground.

3.5.2 Antenna installation site characteristics. The antenna shall be designed such that, with the tower and poles remaining precisely plumb (vertical) and without constraint as to the orientation of the array, installation site grading will not be required unless:

(a) In an area 8 feet (2.438 m) wide and of length bounded by the tower and front pole, the grade from tower to pole exceeds 2 percent, or the roughness of the path exceeds ±18 inches (457.2 mm) from grade.
(b) In the area exclusive of that defined in 3.5.2 (a), above, and within the bounds of the guy anchor points:

(1) The grade exceeds 5 percent in any direction or
(2) The ground roughness exceeds ±4 feet (1.219 m) from grade.

3.5.2.1 The antenna shall remain within the maximum voltage standing wave ratio (VSWR) requirements specified in 3.6.9 when installed over any ground ranging from flat and level to the limits specified in 3.5.2. The pattern requirements specified in 3.6.4 are permitted to vary, since the pattern specifications apply only to flat, level ground.

3.5.2.2 Materials. All additional materials, such as longer radiating element extenders (drops), longer catenaries and guy assemblies, additional anchors and additional attachment hardware, which are necessary to allow antenna installation in any direction and over any site ranging from flat and level to the limits specified in 3.5.2 shall be supplied.

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3.6 Electrical performance. The antenna, with matching unit installed, shall comply with the electrical design requirements of MIL-A-28768 and the performance requirements specified in 3.6.1 through 3.6.10. Unless otherwise specified herein, far field conditions and perfectly conducting ground (infinite conductivity) shall be assumed. For the quad configuration, all requirements except isolation shall be met for each array independently.

3.6.1 Power. The antenna shall be capable of handling the power as classified in MIL-A-28768.

3.6.2 Frequency range. The antenna shall be broadbanded; that is, capable of operating without tuning or adjustment over the entire frequency range of the type specified (see 1.2).

3.6.3 Gain. The vertically polarized antenna gain at the beam maximum shall not be less than +10.5 dB with respect to an isotropic radiator in free space.

3.6.4 Radiation pattern.

3.6.4.1 Elevation. The upper half power (-3 dB) point in the vertical plane which passes through the point of maximum gain shall lie between 20 degrees and 30 degrees elevation (that is, 20 to 30 degrees up from horizontal). The point of maximum gain in this plane shall be at the zero degree elevation angle.

3.6.4.2 Azimuth. The azimuth half power (-3dB) beamwidth shall not be less than 110 degrees nor greater than 165 degrees at the zero degree elevation angle (that is, in the horizontal plane). The beam shall be symmetrical, with the point of maximum gain on the axis of the array.

3.6.4.3 Side and back lobes. The power in any lobe except the one containing the point of maximum gain (main lobe) shall be at least 13 dB below the main lobe maximum at corresponding frequencies.

3.6.5 Polarization. The antenna shall be vertically polarized to the extent necessary to comply with 3.6.3. Cross polarization in the direction of maximum gain shall be at least 20 dB below the vertical component.

3.6.6 Isolation. For quad configurations, the radio frequency isolation between all two array combinations shall be at least 15 dB.

3.6.7 Efficiency. The efficiency of the antenna as defined in MIL-A-28768 shall not be less than 80 percent, including the effects of VSWR. A feedline terminating resistor shall not be used.

3.6.8 Common mode. The antenna shall have a common mode rejection of (that is, be balanced within) at least 15 dB.

3.6.9 VSWR. The VSWR shall not exceed 2.0 to 1 with respect to 50 ohms for all antenna types, classes, configurations and powers.

3.6.10 Noise figure. The antenna shall not contribute more than 1.5 dB of noise relative to kT_oB (Boltzmann's constant x 290°K x bandwidth).

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3.7 Ground system. The antenna shall not be dependent upon an artificial ground system in order to comply with the requirements of this specification. Hence, such a ground system shall not be supplied unless it is requested specifically in the procurement document (see 6.2).

4. QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance provisions shall be in accordance with MIL-A-28768 and as specified herein.

4.2 First article inspection. In addition to the inspection requirements of MIL-A-28768 the following inspections shall be conducted:

(a) Verification of configuration and size requirements of 3.1.1 and 3.3 1/
(b) Verification of wind and ice loading capability, by class, required by 3.4 2/
(c) Verification of compliance with adjustment capability and installation site requirements of 3.5.1 and 3.5.2. 3/
(d) Verification of compliance to the requirements of 3.6.5 through 3.6.8 and 3.6.10 2/

4.3 Inspection procedures. The following inspection procedures supplement those specified in MIL-A-28768.

4.3.1 Gain and pattern measurements. Gain and pattern measurements shall be made (when a fractional scale model is specified) using a standard linearly polarized half-wave dipole, and expressed relative to an isotropic source.

5. PREPARATION FOR DELIVERY

5.1 Preparation for delivery shall be as specified in MIL-A-28768.

6. NOTES.

6.1 Intended use. The antennas covered by this specification are intended for general purpose point-to-point communication using ground wave for short distances and the ionosphere for long haul circuits. They will be permanently installed at world-wide locations, and hence subject to environmental extremes.

6.1.1 Class B is assumed in this specification. Class B is intended to cover generally encountered service conditions. If unusually light or unusually severe wind and ice conditions are known to exist, the class should be specified accordingly (see 6.2).

1/ This requirement shall be demonstrated during installability test.
2/ This requirement may be demonstrated by engineering analysis.
3/ This requirement shall be demonstrated during installability test by visual inspection and by using geometry.

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6.2 Ordering data. In addition to the items specified in MIL-A-28768, procurement documents should specify the following:

(a) Type and configuration (see 1.2)
(b) Class, if other than class B is required (see 3.4)
(c) Artificial ground system, if desired (see 3.7)

6.3 Other notes. The notes contained in MIL-A-28768 are applicable to this specification.

Preparing activity:
Navy - EC

Project 5985-N014

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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